

Medallion Architecture Data Layers

A clean reference for Bronze, Silver, and Gold responsibilities

Flow of value

Land raw data, standardize it, then publish trusted business-ready outputs for analytics and reporting.



Bronze Layer

Capture exactly as received

PURPOSE

Raw, unprocessed source data kept as-is for traceability, replay, and debugging.

OBJECT / LOAD

- Tables
- Full load: truncate and insert

TRANSFORMATIONS

- None - preserve original shape and values

MODELING

None - source-aligned structure.

PRIMARY USERS

- Data engineers
- Data analysts



Silver Layer

Clean and conform the data

PURPOSE

Intermediate layer that prepares standardized, reliable data for downstream analysis.

OBJECT / LOAD

- Tables
- Full load: truncate and insert

TRANSFORMATIONS

- Cleaning and standardization
- Normalization and derived columns
- Enrichment and integration

MODELING

None - usually still close to operational entities.

PRIMARY USERS

- Data engineers
- Data analysts



Gold Layer

Publish business-ready outputs

PURPOSE

Curated data products consumed by reporting, dashboards, and analytics users.

OBJECT / LOAD

- Views
- No load for view-based objects

TRANSFORMATIONS

- Business rules and logic
- Integration and aggregation

MODELING

- Star schema
- Aggregated objects
- Flat reporting tables

PRIMARY USERS

- Data analysts
- Business users

Layer Build Workflow

Each layer follows the same engineering loop while producing different outputs



Tip: keep transformations observable and versioned so every business metric can be traced back to its source.

Source System Interview Checklist

Questions to ask before extracting and loading source data

Goal Confirm ownership, business context, technology constraints, load strategy, security, and operational impact before ingestion starts.

Business Context & Ownership

- ☐ Who owns the data?
- ☐ Which business process does it support?
- ☐ Is system and data documentation available?
- ☐ Is there a data model or data catalog?

Architecture & Technology Stack

- ☐ How is the data stored? Examples: SQL Server, Oracle, AWS, Azure.
- ☐ Which integration patterns are supported? Examples: API, Kafka, file extracts, direct database access.

Extract & Load Requirements

- ☐ Should the pipeline use incremental loads or full loads?
- ☐ Are there data volume limitations?
- ☐ What auth is required? Tokens, SSH keys, VPN, IP allowlisting.
- ☐ What is the expected extract size?
- ☐ How can the source system performance be protected?